

RESEARCH & DEVELOPMENT REVIEW

Sucrose-Assisted Solution Combustion Synthesis of a Multicomponent Metal-Oxide Nanofertilizer

A safety-gated synthesis strategy, suspension-development plan, urea integration pathway, plant-use protocol, and annotated evidence review

Target precursor system

Mg, Fe, Zn, Cu, Ca and Mn nitrates • sucrose fuel • ammonium nitrate considered separately

Prepared as an independent technical review and experimental planning document

Executive conclusion

RECOMMENDED DEVELOPMENT ROUTE Prepare and validate the metal-oxide composite without added ammonium nitrate; quantify composition and dissolution; formulate it as a redispersible suspension; store urea separately and combine only at application during early trials. A coated-urea product should be a later, separately optimized formulation.

The proposed precursor set can produce a nanostructured multicomponent oxide, but the result cannot be assumed to be a physical mixture of MgO, Fe₂O₃, ZnO, CuO, CaO and MnO₂. In one-pot SCS, Fe and divalent metals can form ferrites or spinel solid solutions; manganese oxidation state is atmosphere- and thermal-history-dependent; calcium may remain as CaO, CaCO₃, calcium ferrite, or an amorphous phase. Therefore, the word nanofertilizer is justified only after phase, size, elemental-release, plant-response, and safety evidence are obtained.

There is no universal ideal ratio. Crop, soil pH, baseline nutrient status, irrigation chemistry, application route, particle dissolution, and regulatory limits determine the acceptable dose. The provisional cation design below is a starting point for screening, not a commercial formula. Copper is intentionally kept very low because the distance between deficiency correction and toxicity can be narrow.

CRITICAL SAFETY BOUNDARY Metal nitrates, sucrose, and especially ammonium nitrate form an energetic oxidizer-fuel system. This document does not provide batch masses, an ignition temperature, or a scale-up recipe. Work requires institutional approval, compatible equipment, remote operation, calorimetry, ventilation, and a process-hazard review. Never heat a sealed or confined nitrate-fuel mixture.

1. Scope, research questions, and claims

This review addresses four linked questions: how to develop the composite by SCS; how to decide whether the product is truly nanoscale and agronomically usable; how to formulate a 20 wt% suspension; and whether to apply it alone, with urea, or as a coating. It deliberately distinguishes material synthesis from fertilizer validation. Successful combustion is not evidence of fertilizer efficacy.

- Primary product claim to test: a redispersible, compositionally controlled micronutrient/secondary-nutrient oxide suspension that improves nutrient status without phytotoxicity.

- Secondary claim to test: point-of-use co-application with urea improves nitrogen-use efficiency or biomass relative to equal nutrient doses applied separately.
- Claims not yet supported: controlled release, superior yield, environmental safety, foliar suitability, or “nano” status of every phase.

2. Chemical design of the multication composite

2.1 Expected phase chemistry

The target is chemically more complicated than a premix of isolated oxides. Fe(III) favors spinel ferrites with Mg, Zn, Cu and Mn. A single broad XRD pattern may conceal multiple nanocrystalline phases; an apparently single phase can also have nonuniform elemental distribution. Calcium is less easily incorporated into ordinary spinel ferrites and may segregate. Post-combustion exposure to air and carbon dioxide can carbonate CaO. Because nutrient release depends on phase identity, the phase assemblage is a formulation variable, not merely a characterization result.

Precursor	Likely roles after SCS	Main agronomic function	Key risk/uncertainty
Mg nitrate	MgO and/or Mg ferrite	Mg for chlorophyll and enzymes	Low dissolution in alkaline media; phase locking
Fe nitrate	Fe ₂ O ₃ /Fe ₃ O ₄ and mixed ferrites	Fe for chlorophyll synthesis/electron transfer	Very low availability in high-pH soil
Zn nitrate	ZnO and/or Zn ferrite	Zn-dependent enzymes and growth regulation	Dose-dependent phytotoxicity
Cu nitrate	CuO and/or Cu ferrite	Redox enzymes, lignification	Narrow safety margin; antimicrobial effects
Ca nitrate	CaO/CaCO ₃ /Ca ferrite or amorphous Ca phase	Cell walls, membranes, signaling	Carbonation; high-pH reactivity
Mn nitrate	MnO _x and/or Mn ferrite	Photosystem II and enzymes	Oxidation state strongly process-dependent
Sucrose	Fuel/complexing carbon source; gas generation	No intended nutrient claim after complete combustion	Residual carbon if fuel-rich
Ammonium nitrate	Additional oxidizer/gas former	Nitrogen only if retained, which SCS does not ensure	Major energetic severity; no target metal

2.2 Provisional cation screening design

A conservative screening center can be defined on a metal-cation molar basis, not on nitrate-salt mass and not on oxide mass. One defensible center point is Mg 42%, Ca 35%, Fe 15%, Zn 5%, Mn 2.5%, and Cu 0.5% of total metal moles. This is not an “ideal ratio”; it is a hypothesis that retains Mg/Ca as the bulk, adds Fe, and limits Cu. Hydrate waters and nitrate content must be accounted for separately in any stoichiometric calculation.

Factor	Low	Center	High	Purpose
Fe fraction	8 mol%	15 mol%	25 mol%	Tests ferrite formation vs. discrete oxides
Zn fraction	2 mol%	5 mol%	8 mol%	Tests Zn delivery and toxicity window
Mn fraction	1 mol%	2.5 mol%	5 mol%	Tests Mn response and phase changes
Cu fraction	0.1 mol%	0.5 mol%	1.0 mol%	Defines minimum effective, non-toxic Cu
Mg:Ca balance	55:45	55:45	70:30	Tests solubility and secondary-nutrient balance

Hold total metal dose constant in plant trials. Changing the cation composition and total solids at the same time makes efficacy impossible to interpret. Each formulation should be compared with soluble ionic controls delivering the same Mg, Fe, Zn, Cu, Ca and Mn amounts.

3. SCS chemistry and stoichiometric framework

SCS couples nitrate oxidizers with an organic fuel. Sucrose coordinates metal ions imperfectly compared with classic chelating fuels, but it can promote homogeneous mixing and generates a porous product through rapid gas evolution. The fuel-to-oxidizer equivalence ratio (often denoted ϕ) controls combustion mode, peak temperature, residual carbon, crystallinity, particle necking, and phase selectivity. Literature repeatedly shows that fuel identity and amount are not transferable constants across different nitrate hydrates or cation mixtures.

For planning, calculate an oxidizing-valence total from nitrate groups and a reducing-valence total from the carbon and hydrogen in sucrose, using the actual certificate-of-analysis formula for every hydrated salt. Define ϕ consistently in the laboratory worksheet, document whether ϕ = reducing/oxidizing or its reciprocal, and have the balance independently checked. The candidate window must then be narrowed by DSC/TGA and, for scale-up, adiabatic calorimetry. Do not infer safety from a published formula for a different oxide.

AMMONIUM NITRATE DECISION Exclude added ammonium nitrate from the initial SCS matrix. The metal nitrates already provide oxidizing nitrate. If ammonium nitrate is scientifically necessary, treat it as a separate hazard-controlled factor only after calorimetry and formal process-safety approval; do not assume it improves

nutrient value after combustion.

4. Recommended stage-gated synthesis program

Stage 0 — Define the target product profile

1. Specify crop, growth stage, soil or substrate, pH, baseline nutrient analysis, application route, and intended nutrient dose per hectare or per pot.
2. Define measurable acceptance criteria: elemental composition, phase identity, median agglomerate size, redispersibility, dissolution, nitrate residue, microbial response, and plant safety.
3. Create a batch record that distinguishes nitrate hydrate formula, assay, lot, water content, and metal-mole contribution.

Stage 1 — Establish non-energetic analytical controls

4. Prepare physical mixtures of the already available oxide nanopowders at the same cation ratios. These controls reveal whether SCS-created phases improve or reduce nutrient release.
5. Prepare soluble nitrate controls for plant studies at equal elemental doses; these distinguish a nano/phase effect from ordinary ion nutrition.
6. Run compatibility tests between each oxide, water, candidate dispersant, urea solution, and irrigation water before combustion work.

Stage 2 — Single- and binary-cation SCS controls

7. Develop Mg, Fe, Zn, Cu, Ca and Mn single-cation products independently under an institution-approved microscale screening plan.
8. Study the most probable mixed phases: Mg–Fe, Zn–Fe, Mn–Fe, Cu–Fe, and Ca–Fe. Use XRD/Rietveld analysis and ICP digestion to close the mass balance.
9. Use DSC/TGA of dried precursor gels to compare onset, heat release pattern, residual mass, and multistep decomposition. Stop any formulation with abrupt or poorly controlled exothermicity.

Stage 3 — Multication design of experiments

10. Select a small factorial or response-surface design around the cation center point. Keep total metal concentration and precursor water basis controlled.

11. Treat fuel equivalence, solids concentration before combustion, and post-treatment severity as independent process factors. Do not change all three in one experiment.
12. Perform reactions only in equipment and scale authorized by the hazard review: open, vented, heat-resistant containment; local exhaust; remote observation; no confinement; no combustible surroundings.
13. Quarantine each powder until residual oxidizer, free carbon, phase composition, and thermal stability are known.

Stage 4 — Post-processing and release testing

14. Choose post-heat-treatment conditions from TGA/XRD rather than copying a universal calcination temperature. Excessive treatment can sinter particles and suppress nutrient release.
15. If washing is used, measure metals in every wash to detect preferential leaching and composition drift. Drying can create hard agglomerates; compare gentle drying and redispersion strategies.
16. Accept a batch only when replicate composition, phase, BET area, microscopy, and dissolution fall within predeclared ranges.

5. Minimum characterization package

Question	Method	Decision criterion
What phases formed?	Powder XRD with Rietveld refinement; Raman where useful	No unidentified dominant phase; replicate phase fractions
Is it nanoscale?	TEM with statistically adequate particle counts; SEM for agglomerates	Report primary particles and agglomerates separately
Is composition correct?	Total digestion ICP-OES/ICP-MS; XRF as screening	Metal recovery and batch homogeneity within validated limits
What oxidation states exist?	XPS; Mn/Fe-sensitive spectroscopy when needed	Oxidation-state assignment tied to dissolution and plant response
Does it redisperse?	DLS/laser diffraction, zeta potential, rheology, sediment height	Stable or readily redispersible over intended shelf life
How fast do nutrients release?	Time-resolved dissolution in water, soil extract, and root-zone simulant	Element-specific release curve; no burst Cu
Is oxidizer/fuel residue present?	Ion chromatography for nitrate; CHN/TOC; DSC/TGA	Below internal acceptance limit; no residual exotherm
Is it biologically safe?	Seed germination, root elongation, soil respiration, earthworm/microbial screens	No adverse effect at intended and margin doses

6. Developing the 20 wt% suspension

A 20 wt% suspension is a concentrated stock, not automatically a field-use concentration. Metal-oxide nanoparticles commonly aggregate because of high surface energy, multivalent-ion bridging, and high ionic strength. Urea and fertilizer salts can compress the electrical double layer and destabilize a suspension that appears stable in deionized water.

17. Measure true solids content by drying a representative aliquot and confirm elemental concentration by ICP. Dose plants by mg of each element, not by an assumed powder label.
18. Screen 5, 10, 15 and 20 wt% solids. Record viscosity, yield stress, sedimentation, redispersibility, particle-size distribution and zeta potential at day 0, 1, 7, 14 and 28.
19. Screen only agriculturally acceptable dispersants at low levels, such as citrate, lignosulfonate or a suitable polycarboxylate, after checking metal complexation, phytotoxicity and local registration status. Select by data, not by visual appearance alone.
20. Use controlled high-shear mixing and, if appropriate, validated sonication during development. Avoid aerosol generation. Define an order of addition and energy input that can be reproduced at scale.
21. Test the actual dilution water. Hardness, phosphate, carbonate and pH can drive precipitation or heteroaggregation. Include nozzle-screen and drip-emitter tests if application equipment is relevant.

STORAGE RULE Store the nanocomposite stock separately from urea during early development. Prepare a fresh, diluted tank mix immediately before use only after a jar test confirms no flocculation, gas generation, pH excursion, or nozzle blockage.

7. Independent use, urea co-application, and coating

Strategy	Scientific advantage	Main limitation	Recommendation
Separate application	Maximum control of N and micronutrient timing; simplest attribution	Two application operations	Best for first efficacy trials
Fresh tank mix with urea	One application; tests synergy at equal doses	Ionic-strength aggregation; pH and storage instability	Best practical development option after jar tests
Pre-blended liquid concentrate	Convenient product concept	High risk of sedimentation, complexation and microbial/storage change	Do not pursue until stability data exist
Nanocomposite-coated urea	Potential coordinated N + micronutrient delivery	Requires coating equipment, binder, abrasion/dust and release	Later product-development stage

Strategy	Scientific advantage	Main limitation	Recommendation
validation			

7.1 Point-of-use mixture with urea

Prepare urea solution in the bulk water first, verify complete dissolution and temperature recovery, then add a pre-diluted nanocomposite suspension under agitation. Use the mixture promptly. The laboratory should define a maximum hold time from particle-size and sedimentation data. Compare the same nutrient doses applied together and separately; otherwise an apparent synergy may simply reflect a higher nutrient dose.

7.2 Coated-urea development

Published ZnO-coated urea studies demonstrate feasibility, but they do not validate this six-metal composite. Begin with low coating loadings in a research design, for example 0.25%, 0.5% and 1.0% composite by urea mass, and include uncoated urea, binder-only urea, and soluble-micronutrient controls. These are screening levels, not use recommendations. Select an agriculturally acceptable binder, apply in contained coating equipment, dry without nitrate-fuel contact, and measure coating uniformity, attrition, dust, crushing strength, dissolution and N release. The SCS powder must contain no residual oxidizer or fuel before any contact with urea granules.

8. Plant-use protocol

Soil or substrate drench is the preferred first route. It reduces inhalation and leaf-burn risk and allows the root zone to buffer release. Foliar use should be a separate study with much lower exposure, droplet/nozzle characterization, leaf residue analysis and food-crop risk assessment.

8.1 Greenhouse dose-finding design

Treatment	Purpose
Water/vehicle control	Baseline growth and dispersant effect
Recommended conventional fertilizer	Agronomic benchmark
Urea alone	Nitrogen benchmark
Soluble metal-ion mix + urea	Equal-element ionic control
Physical oxide premix + urea	Tests effect of SCS-created phases

Treatment	Purpose
SCS composite alone	Separates micronutrient effect from N
SCS composite + urea, separate timing	Tests timing and interaction
Fresh SCS composite/urea tank mix	Tests operational co-application

For the first range-finding study, a cautious total-composite series such as 5, 10, 25 and 50 mg/L in the applied solution can identify gross toxicity, but the scientifically meaningful quantities are the resulting mg/L and mg/pot of Cu, Zn, Mn, Fe, Mg and Ca measured by ICP. These concentrations are research starting points, not field recommendations. Include at least one higher “margin” dose only under controlled greenhouse containment, and stop escalating when root elongation, chlorophyll, biomass or soil respiration declines.

- Primary endpoints: germination, root length, shoot/root dry mass, chlorophyll index, visual injury, tissue nutrient concentration, and yield component appropriate to the crop.
- Mechanistic endpoints: photosynthetic efficiency, antioxidant enzymes, oxidative-stress markers, rhizosphere pH, available soil nutrients, microbial respiration, and residual metals in soil.
- Nanoparticle fate: dissolved vs. particulate fraction, tissue localization where justified, and mass balance among plant, soil and leachate.

Advance only a dose that improves at least one agronomic endpoint without statistically significant injury, excessive tissue Cu/Zn/Mn, microbial suppression, or unacceptable soil accumulation. Confirm in a second soil type and then in a field plot under local regulatory oversight.

9. Interpretation: how the composite could help plants

Potential benefit arises from corrected deficiencies and controlled ion release, not from “nano” size by itself. Mg supports chlorophyll; Fe supports electron transport and chlorophyll formation; Zn supports enzymes and growth regulation; Mn participates in photosystem II and antioxidant enzymes; Cu supports redox enzymes and lignification; Ca supports cell walls, membranes and signaling. A porous high-area oxide may dissolve gradually or interact strongly with roots. However, ferrite formation can also lock nutrients into sparingly soluble phases, while nanoscale surfaces can generate reactive oxygen species. The same material can stimulate growth at a low dose and inhibit it at a higher dose—a hormetic response commonly reported for metal-oxide nanoparticles.

The strongest evidence of fertilizer action is therefore a chain: characterized material → measured dissolution/speciation → improved plant nutrient status → improved growth or yield → acceptable environmental fate. Biomass alone is insufficient because stress can transiently alter growth and because antimicrobial particles may change the rhizosphere.

10. Decision gates and failure modes

Gate	Pass evidence	Typical failure response
Thermal safety	No uncontrolled exotherm in approved calorimetric window	Reduce scale; change composition/fuel; remove AN
Material identity	Reproducible phases, composition and particle statistics	Return to binary controls and process DOE
Suspension quality	Redispersible, dose-uniform, application-compatible	Change solids level/dispersant/mixing order
Nutrient release	Measurable, balanced release without Cu burst	Adjust phase composition or abandon one-pot route
Plant safety	No injury at intended dose plus safety margin	Lower dose; separate elements; change route
Agronomic efficacy	Benefit vs equal-element soluble and physical-mixture controls	Do not claim nano advantage
Environmental acceptability	No unacceptable soil/leachate/microbial effect	Reformulate or stop development

11. Direct evidence: nano metal-oxide composite fertilizers

This section uses a strict inclusion rule: the material must contain a metal-oxide nanophase combined with another oxide, dopant, mineral, carbon, silica, or fertilizer carrier, and it must have been tested on plants or as a nutrient-delivery fertilizer. SCS-only papers, single free oxides, photocatalysis studies, and general reviews are excluded. Under this definition, the primary literature is limited; no published plant study located here tests the exact Mg–Fe–Zn–Cu–Ca–Mn oxide composite. The small evidence base is a research gap, not a reason to add unrelated citations.

- Ingavale et al. (2026), Zn- and Cu-doped MnFe₂O₄ nanofertilizer in fenugreek.** The closest chemical analogue: a spinel ferrite carrying Mn and Fe with Zn and Cu dopants. The study reports concentration-dependent improvements in fenugreek germination, chlorophyll and growth, while reinforcing the need to optimize dose. [Link](#)

2. **Durgude et al. (2022), mesoporous-silica and graphene-supported FeO/ZnO nanocomposites in rice.** Foliar Fe/Zn nanocomposites increased Fe and Zn uptake and grain fortification. The reported mesoporous-silica treatment containing 30 ppm Zn plus 5 ppm Fe improved grain yield by 53% over conventional fertilization in that experiment. [Link](#)
3. **Farhangi-Abriz and Ghassemi-Golezani (2021), biochar-based Mg/Mn oxide nanocomposites in safflower.** Biochar-supported metal-oxide nanocomposites improved soil properties and safflower root/shoot growth under salinity, reduced sodium-related injury, and improved Mg/Mn nutrition. [Link](#)
4. **Ghassemi-Golezani et al. (2021), combined biochar-MgO and biochar-MnO nanocomposites.** In salt-stressed safflower, the combined BNC-MgO and BNC-MnO treatment was the strongest treatment for photosynthetic performance, tissue nutrient status, reduced sodium accumulation and lower oxidative stress. [Link](#)
5. **Ghassemi-Golezani and Farhangi-Abriz (2021), biochar-based Mg/Mn oxide nanocomposites and safflower productivity.** A related crop experiment reported improved root development and productivity under salt stress, supporting a carrier-bound Mg/Mn oxide strategy rather than an unbound six-powder suspension. [Link](#)
6. **Jafarirad et al. (2021), MnO₂/perlite nanocomposite in Hypericum perforatum cultures.** MnO₂ nanoparticles approximately 20–50 nm were supported on nano-perlite. Selected concentrations improved shoot-culture growth and altered valuable secondary metabolites; responses were concentration-specific. [Link](#)
7. **Morfín-Gutiérrez et al. (2025), Fe₃O₄@MCM-48 nanofertilizer for tomato.** An iron-oxide/mesoporous-silica composite was evaluated as an iron nanofertilizer in tomato, providing direct evidence for a carrier-supported oxide rather than a freely dispersed oxide mixture. [Link](#)
8. **Ehsan et al. (2022), phytogetic Ag/ZnO bimetallic nanomaterial with fertilizer in wheat.** The Ag/ZnO composite plus fertilizer improved yield-related parameters in the tested wheat varieties. Because Ag is not a plant nutrient and can be antimicrobial, this supports composite efficacy but is not a formulation model for the proposed nutrient-only product. [Link](#)
9. **Ebadollahi et al. (2019), TiO₂/perlite nanocomposite in plant tissue culture.** TiO₂ supported on nano-perlite influenced growth and volatile-compound production in plant cultures. It is relevant to carrier effects but does not establish micronutrient fertilizer value for Mg/Fe/Zn/Cu/Ca/Mn. [Link](#)
10. **Ram et al. (2022), nanosilica/reduced-graphene composite with urea for rice.** A urea-containing nanocomposite platform was evaluated for nitrogen delivery and rice growth/yield.

It supports the feasibility of incorporating a nanocomposite into a urea product, but it is not a multi-metal-oxide formula. [Link](#)

11. Kottegoda et al. (2020), urea–calcium-phosphate hybrid carrying Cu, Fe and Zn. A multinutrient nano-hybrid combined urea with a calcium-phosphate matrix and Cu/Fe/Zn. It is useful evidence for coordinated nutrient delivery, although it is not a pure mixed-metal-oxide composite. [Link](#)

12. Reporting template for each experiment

- Material identity: batch ID, precursor lot and hydrate form, metal-mole design, fuel-equivalence convention, atmosphere, approved equipment, and deviations.
- Thermal evidence: DSC/TGA sample basis, onset(s), residual mass, and calorimetry reference; record that no confinement was used.
- Powder evidence: XRD/Rietveld, TEM particle statistics, SEM agglomerates, BET, ICP mass balance, XPS where needed, residual nitrate and carbon.
- Suspension evidence: actual solids, pH, conductivity, viscosity, DLS/laser diffraction, zeta potential, sedimentation, redispersibility, dilution-water compatibility, and storage time.
- Plant evidence: crop/cultivar, soil analysis, experimental unit, randomization, replication, complete nutrient doses, application route, timing, endpoints, statistics, tissue/soil/leachate mass balance.

13. Final recommendation

The best scientific route is a staged program, not immediate scale-up of the full nitrate–sugar–ammonium-nitrate mixture. Start without added ammonium nitrate, use the provisional cation composition only as a screening center, and allow phase/dissolution evidence to decide whether one-pot SCS is superior to a physical oxide premix. Formulate a characterized powder as a redispersible stock, then use separate storage and fresh co-application with urea for early agronomy. A coated-urea product is justified only after the composite has passed thermal, compositional, release, plant-safety and environmental gates.

INFORMATION STILL REQUIRED FOR A FIELD RECOMMENDATION Target crop and cultivar; soil pH and test results; irrigation-water chemistry; application route; desired N rate; local fertilizer/nanomaterial regulations; and measured ICP composition and dissolution of the actual SCS batch.